# The Influence of Cultural Factors in Software Quality Assessment Models A Systematic Literature Review

1st Ni Nyoman Utami Januhari
*Dept. of Informatics Doctorate*
*Universitas Amikom Yogyakarta*
Yogyakarta, Indonesia
utamijanuhari@students.amikom.ac.id

2nd Arief Setyanto
*Dept. of Informatics Doctorate*
*Universitas Amikom Yogyakarta*
Yogyakarta, Indonesia
arief@amikom.ac.id

3rd Kusrini
*Dept. of Informatics Doctorate*
*Universitas Amikom Yogyakarta*
Yogyakarta, Indonesia
kusrini@amikom.ac.id

4th Ema Utami
*Dept. of Informatics Doctorate*
*Universitas Amikom Yogyakarta*
Yogyakarta, Indonesia
ema.u@amikom.ac.id

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***Abstract*—This study focuses on the influence of cultural factors within software quality assessment models through a Systematic Literature Review (SLR) approach. The primary goal of this research is to identify and analyze the distribution of publications from 2019 to 2024 in the field of software quality research, particularly quality assessment models, and to assess the cultural factors frequently considered within these models. The methodology used includes defining research questions, inclusion and exclusion criteria, comprehensive literature search, critical assessment of included studies, data extraction and management, as well as analysis and interpretation of results. The findings indicate that cultural factors have a substantial impact on the development and assessment of software quality, highlighting the importance of integrating diverse behavioral norms and interaction styles in effective software development. Based on these findings, the recommendations proposed include the development of assessment methods that can more effectively accommodate cultural diversity, as well as the integration of cultural aspects into existing software quality models.**

***Keywords—software quality models, quality assessment models, cultural factors in software quality, systematic literature review of software quality***

## I. INTRODUCTION

The research pattern on the influence of cultural factors on software quality development has become a research trend in recent years because cultural factors can influence software development outcomes, indicating the need for further integration of development team characteristics and organizational culture into software quality assessment models [1],[2]. Cultural and geographical dispersion in software development impacts productivity, making the integration of cultural factors in software quality assessment essential for enhancing user satisfaction and providing more [3],[4],[5].

The root of the problems and difficulties in integrating cultural factors into quality assessment models often stem from standard approaches that do not consider cultural diversity which can affect quality assessment [6]. This emphasizes that a deep understanding of the culture and social needs of users is essential for the creation of successful and lasting software systems [7]. To improve software quality, there needs to be a deeper understanding of how cultural factors influence quality assessments. By identifying the gaps that exist in current software quality assessment models, this study aims to deepen the knowledge of the role of cultural factors in software quality assessment models and serve as a reference for the development of models that are more relevant and sensitive to cultural factors.

The main objective of the systematic review is to identify and analyze the distribution of publications in 2019 - 2024 in the field of software quality research, specifically software quality assessment models, determine aspects of the research the most frequently researched, and perform a comparison between quality characteristics based on research results. This SLR also aims to examine cultural factors that are often considered in software quality models, assess the types of software that require quality assessment with consideration of cultural factors, and explore how these factors influence the development and assessment of software quality. Additionally, this SLR aims to identify key challenges in software quality development research.

## II. METHODOLOGY

This study employs the Semantic Literature Review (SLR) methodology to examine research related to the influence of cultural factors on software quality assessment. It consists of five stages: planning, literature search, inclusion and exclusion criteria determination, data collection, and data analysis. The planning stage defines the research strategy, while the literature search gathers relevant information. The inclusion and exclusion criteria ensure that only pertinent studies are analyzed. The selected data is then collected and analyzed to identify trends and the latest technological advancements in fact verification. This methodology provides a comprehensive overview of the recent progress and challenges in this field.

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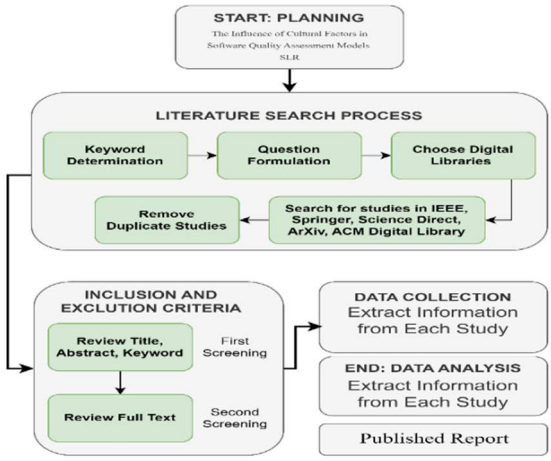



Fig. 1. Literature search and selection

### A. Literature Search Strategies

A comprehensive publication search was carried out using databases, through the digital library portals of four major publishers: IEEE Xplore & Access, Springer, Science Direct, and ACM Digital Library. These keywords are: ("Software quality models" OR "Quality assessment models" "Software Quality" OR "Standards" OR "Metrics") AND "cultural factors in software quality".

### B. Research Questions

The research question was constructed using the criteria of PICOC (Population, Intervention, Comparison, Outcome, Context). Table I shows the formulation of the research questions built from PICOC.

TABLE I. PICOC BASED RESEARCG QUESTION FORMULA

| PICOC | Question Formula |
|---|---|
| Population | Software quality assessment model |
| Intervention | Quality, assessment, method, model |
| Comparison | Model comparison, identification of cultural factors |
| Outcome | The influence of cultural factors |
| Context | Software, engenering |

Using the PICOC framework, researchers can develop clear research questions, establish inclusion and exclusion criteria, and systematically guide literature searches to examine the influence of cultural factors on software quality assessment. The research questions are detailed in:

TABLE II. TABLE TYPE STYLES RESEARCH QUESTION

| | |
|---|---|
| RQ1 | What is the distribution of publications by year in software quality research? |
| RQ2 | How to compare software quality characteristics based on research results? |
| RQ3 | What cultural and cross-cultural criteria are the most studied in software quality research? |
| RQ4 | What cultural factors are most often considered in software quality models? |
| RQ5 | What types of software require quality assessment from cultural factors? |
| RQ6 | What are the influences of cultural factors in software quality assessment models? |
| RQ7 | What are the main challenges in software quality development research? |

### C. Selection Criteria

Previous studies have evaluated Quality Assessment (QA) and software quality metrics from 2004 to 2019 [4], [8]. This research focuses on the impact of culture on quality assessment from 2019 to 2024. Inclusion criteria are publications from 2019-2024 that discuss quality assessment models with cultural factors, written in English. Exclusion criteria include publications before 2019, irrelevant articles, or those not peer-reviewed. The selection process involves an initial screening based on titles and abstracts from 314 articles, followed by full-text screening, resulting in a final selection of 47 articles.

### D. Data Extraction and Management

Data Extraction: Identify and extract data such as author names, publication year, assessment model used, and key results from each study. Data Management: Use reference management software or spreadsheets to organize the extracted data.

### E. Analysis and Interpretation of Results

Data Analysis: Categorize the results based on major themes or topics, such as the influence of teams on software quality, cultural adaptation in quality models, and research challenges. Interpretation: Compare study results with existing literature, identifying trends, gaps, or areas that require further research.

### F. Published Report

The report includes an introduction, methodology, results, discussion, and conclusion. The results are presented with tables and charts for ease of understanding, and the publication is submitted to a relevant journal.

## III. RESULTS

### A. RQ1: What Is The Distribution Of Publications By Year In Software Quality Research?

The distribution of software quality research publications varies each year. In 2023, the highest number of publications was recorded with 13 publications on team-based quality models, quality assessment, and organizational culture in digital innovation [1], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]. In 2021, there were 11 publications discussing quality requirements, team dynamics, and ethics [21], [22], [8], [23], [24], [7], [25], [26], [27], [6], [47]. In 2020, 9 publications focused on quality assessment, organizational culture, DevOps, and human factors [12], [13], [14], [15], [16], [17], [18], [19], [20]. In 2024, 7 publications addressed quality assurance, sustainability, and architectural support [42], [2], [43], [44], [45], [46], [3]. Fewer publications were recorded in 2022 (4) and 2019 (3), covering topics such as quality models, measurement, and software sustainability [4], [5], [28], [29], [9], [10], [11].

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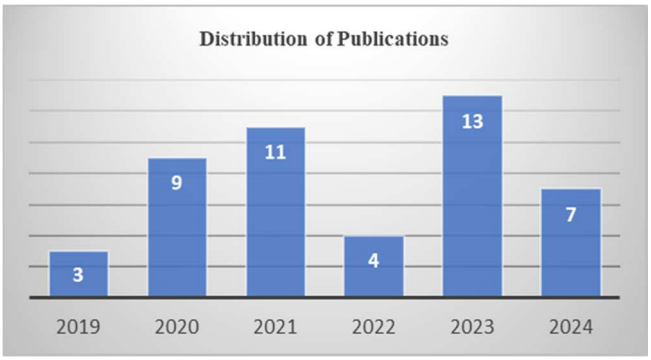



Fig. 2. Distribution of publications

### B. RQ 2: How To Compare Software Quality Characteristics Based On Research Results?

Based on Table 2, the most frequently mentioned software quality characteristic across various models is usability, appearing in 28 models, followed by reliability in 25 models, ease of use in 24 models, and efficiency and portability, each appearing in 23 models. These characteristics reflect the key aspects considered in software quality evaluation, highlighting the importance of usability, reliability, ease of use, efficiency, and portability in ensuring that software can operate effectively in diverse environments and usage conditions. Several models that encompass usability and reliability, which are related to ease of use, are outlined in: ISO/IEC 25010, McCall's model, Boehm's model, and GQM (Goal Question Metric), which support the integration of dynamic and relevant aspects in software quality evaluation, [31], [11].

TABLE III. SOFTWARE QUALITY CHARACTERISTICS

| Software Quality Models comparison of software quality characteristics | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Aljarallah & Lock 2019, | | | | | | | | | | G. L. Saini et al. 2020, | | | | | | | Lloy Pinedo et al. 2023 | | | | | | | | | | | | | |
| No | Characteristics | McCall's | Boehm's | Evans& | Deutsch & | FURPS (+) | Dromey's | ISO 9126 | ISO 25010 | IEEE | AOSQU AMC | Bowen | Murine | Kazman | Khosravi K. | Ghezzi | SATC's | QMOOD 200 | AOSQ 2012 | CSDQ 2009 | DEQUALITE | UML | Gillies | OOQM | Perry | Bertoa | Alvaro | Rawashdeh | Sharma A | Kumar | SEI 1995 | SPARDAT | Total |
| 1 | Portability | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | 23 |
| 2 | Reliability | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | 25 |
| 3 | Efficiency | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 23 |
| 4 | Maintainability | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | 24 |
| 5 | Usability | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 28 |
| 6 | Integrity | ✓ | | ✓ | ✓ | | | | | | | ✓ | ✓ | | | ✓ | | | | | | ✓ | ✓ | | ✓ | | | | | | | 9 |
| 7 | Correctness | ✓ | | ✓ | ✓ | | | | | | | ✓ | ✓ | | | | | | | | | | ✓ | | ✓ | | | | | | | 7 |
| 8 | Testability | ✓ | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | ✓ | ✓ | | ✓ | | | | | | | 7 |
| 9 | Flexibility | ✓ | | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | ✓ | | ✓ | | | | | | | 11 |
| 10 | Reusability | ✓ | | ✓ | ✓ | | ✓ | | | | | ✓ | ✓ | | ✓ | ✓ | | ✓ | | | ✓ | ✓ | ✓ | | ✓ | | | | | | | 12 |
| 11 | Interoperability | ✓ | | ✓ | ✓ | | | | | | | ✓ | ✓ | | | | | | | | | | | | | | | | | | | 8 |
| 12 | Understandability | | ✓ | | | | | | | | | | | | ✓ | | | ✓ | | | ✓ | ✓ | | | | | | | | | | 5 |
| 13 | Modifiability | | ✓ | | | | | | | | | | | | | | | ✓ | | | ✓ | ✓ | | | | | | | | | | 4 |
| 14 | Functionality | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | ✓ | ✓ | | | 16 |
| 15 | Performance | | | | | ✓ | | | ✓ | | | | | | | | | | | | | | | | | | | | | | ✓ | | 3 |
| 16 | Supportability | | | | | ✓ | | | | | | | | | | | ✓ | | | | | | | | | | | | | | | | 2 |
| 17 | Compatibility | | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 18 | Security | | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | ✓ | | 2 |
| 19 | Verifiability | | | ✓ | ✓ | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | 3 |
| 20 | Expandability | | | ✓ | ✓ | | | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | | 5 |
| 21 | Safety | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | | 2 |
| 22 | Manageability | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 23 | Survivability | | | | ✓ | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | 2 |
| 24 | Human En | | ✓ | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 25 | Accuracy | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | 2 |
| 26 | Robustness | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | 2 |
| 27 | Scalability | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | 2 |
| 28 | Modularity | | | | | | | | | | | | | | ✓ | | | ✓ | | | ✓ | | | | | | | | | | | | 3 |
| 29 | Feasibilty | | | | | | | | | | | | | | | | ✓ | | | | | | | | | | | | | | | | 1 |
| 30 | Dependability | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | | 1 |
| 31 | Applicability | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | 1 |
| 32 | Adaptability | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | 1 |
| 33 | Evolvability | | | | | | | | | | | | | | | | | | ✓ | | | | | | | | | | | | | | 1 |
| 34 | Learnability | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | | | | | | | | | | | 2 |
| 35 | Simplicity | | | | | | | | | | | | | | | | | | | | ✓ | | | | | | | | | | | | 1 |
| Total Number | | 11 | 7 | 12 | 15 | 5 | 7 | 6 | 8 | 6 | 6 | 13 | 11 | 6 | 7 | 8 | 8 | 8 | 7 | 6 | 12 | 12 | 11 | 6 | 11 | 5 | 5 | 5 | 6 | 4 | 4 | 3 | |

### C. RQ 3: What Cultural And Cross-Cultural Criteria Are The Most Studied In Software Quality Research?

The cultural and cross-cultural criteria most extensively studied in the context of software quality are the various dimensions related to the influence of organizational culture on software development practices. These aspects include organizational culture (quality models, team motivation, and development processes) [10], [12], [13], [21], [8], [14], [29], [32], [3], [44], [45], [46], [18], [37], [19], [39]. Communication and collaboration (team communication, supporting team integration, and managing team heterogeneity) [1], [14], [16], [17], [47], [7], [34], [37]. Practices and processes (development practices and processes, including effort estimation and testing methodologies [5], [21], [40], [17], [25], [43].




Fig. 3. Cultural and cross-cultural criteria in software quality research

### D. RQ 4: What Cultural Factors Are Most Often Considered In Software Quality Models?

Culturally sensitive software quality assessments address key aspects such as: 1) Language and communication [31], [32], [34], [7], [20]; 2) Social and cultural [1], [13], [22], [16],

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[23], [36], [7], [25]; 3) Organizational and team [1], [16], [43], [36], [18], [37]; 4) Technology and development [21], [8], [44]; 5) Regulatory and quality [32], [46]; 6) Behavioral and interaction [42], [18], [37]. Based on RQ4, these factors ensure that the software is aligned with the technical standards and cultural context of its users.

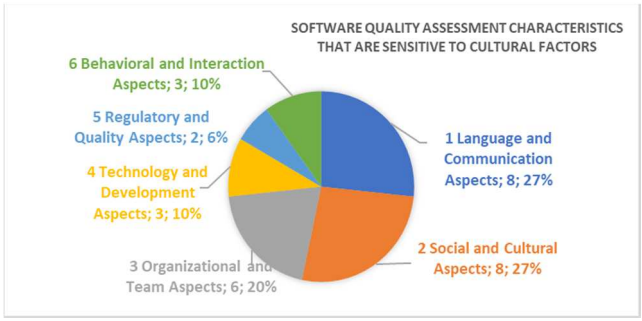



Fig. 4. Software quality assessment characteristics that are sensitive to cultural factors

### E. RQ 5: What Types Of Software Require Quality Assessment From Cultural Factors?

Software that requires quality assessment of cultural factors includes platforms and applications (e-commerce, AI applications, and social platforms) [34], [35], [36], [7], [25], [27]. Characteristics and Features of Multiculturalism (supporting multilingual, multicultural, and regional sensitivity aspects) [35], [44], [25]. Management and information technology (multinational management systems and ERPs to manage international-scale operations and information) [44], [36], [27]. Specific types of software (educational and healthcare software) [46].

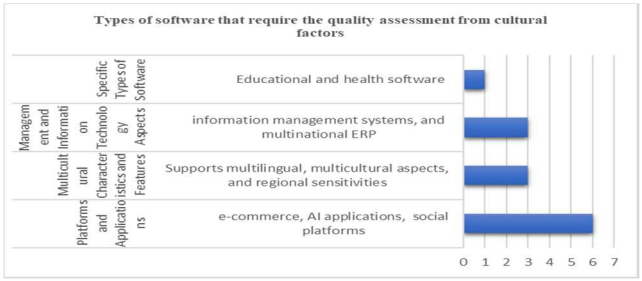



Fig. 5. List of concepts in software quality assessment based on cultural factors

### F. RQ 6: What Are The Influences Of Cultural Factors In Software Quality Assessment Models?

Cultural factors significantly influence software quality assessment within organizations. The key aspects include: 1) Organizational structure and processes, which affect the implementation of quality specifications [21], [9], [10]; 2) Team quality and motivation through reward systems [1], [12], [17]; 3) Cultural practices and norms that prioritize quality [13], [47], [29], [30]; 4) Positive team dynamics and communication [15], [16], [32], [42]; 5) Compliance and innovation, focusing on user satisfaction [40], [31], [18], [37]. By integrating these cultural factors, organizations can develop more effective software quality assessment models that reflect both internal and external expectations.

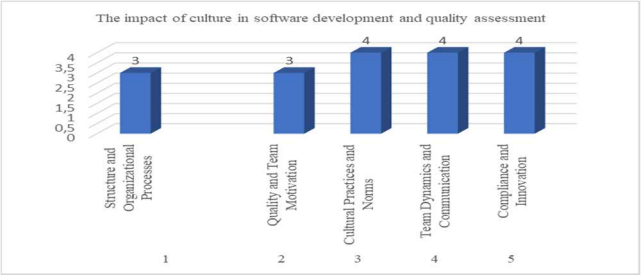



Fig. 6. Cultural factors in software development and quality assessment

### G. RQ 7: What Are The Main Challenges In Software Quality Development Research?

The results of a systematic review of published literature reveal several research gaps, such as challenges in aspects of: 1) quality models and evaluation [12], [13], [4], [8], [40], [31], [34], [42], [43], [23], [36], [24], [25]; 2) Cultural and social factors [12], [13], [4], [15], [8], [34], [43], [44], [23], [36], [46], [24], [18], [20]; 3) Quality integration and management [22], [4], [16], [6], [8]. As well as the gap between research and industry practice, as shown in the figure 7.




Fig. 7. List of concepts and descriptions regarding issues and gaps in software quality development

## IV. Discussion

### A. Interpretation of Findings

A Systematic Literature Review (SLR) from 2019-2024 indicates that 2023 was the peak year for publications on software quality, with usability being the most highlighted characteristic. Cultural factors such as language and social norms significantly influence quality evaluation, especially in global sectors.

### B. Comparison With The Existing

Our study explores the influence of cultural factors on software quality models, differing from other research that typically focuses on development efficiency and user satisfaction [31], [9]. Cultural factors between countries play a role in assessing software quality, team organization, and resource management [21]. Cultures that support innovation and collaboration often produce high-quality software needs [6], [40]. The implementation of cultural elements in processes and standards requires adaptation. For example, in cultures that prioritize compliance, rigorous testing processes are crucial [23], while cultures more open to innovation encourage the creation of innovative software [27]. Quality models and culturally adapted evaluation methods, supported by culturally oriented training, enable companies like those in Japan to adopt additional validation stages, facilitating cultural adaptation and producing high-quality software. [21], [8].

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### C. Strengths and Limitations of the Review

Strengths and Limitations This study introduces a new perspective on the role of cultural factors in software quality and provides an objective and comprehensive review. However, it is limited by a constrained cultural perspective and reliance on available literature, which restricts the strength of conclusions.

## V. Conclusions

This study highlights the importance of incorporating cultural factors into software quality assessment to enhance usability and user satisfaction. It identifies gaps such as the lack of model adaptation to new technologies and cultural variations. To address these gaps, it is recommended to integrate cultural aspects into quality evaluation methods, utilizing frameworks focused on cultural adaptability and sensitivity. Employing Hofstede's dimensions like Power Distance, Individualism vs. Collectivism, and Uncertainty Avoidance is suggested to develop a more inclusive and effective assessment model.

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